



: : Project Work on : :

**INDIAN AGRICULTURE - THE GROWTH
ENGINE OF NATION**

: : Submitted by : :

RAHUL GANGULY

Roll no: BS2337

Course:- Statistical Methods 4

: : Under the guidance : :

Prof. AYANENDRANATH BASU

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Rahul Ganguly

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1 Abstract

Agriculture has been the cornerstone of India's economy and culture for centuries. Often referred to as the backbone of rural India, it not only sustains a large section of the population but also contributes significantly to the country's GDP and employment. Despite rapid industrialization and the growth of the service sector, agriculture continues to play a crucial role in the socio-economic fabric of the country.

India, with its diverse agro-climatic zones and rich agricultural heritage, is home to a wide range of food and non-food crops. The country's agricultural landscape is shaped by varying soil types, climatic conditions, and traditional cultivation practices, resulting in the growth of crops that sustain millions of livelihoods.

In this article our main goal will be :

- i) To identify the states of India which has largest amount of production of different crops (rice,wheat,jute etc.)
- ii) explore any potential correlations between crop yield and different climatic factors(like rainfall , temperature etc)
- iii) To analyze overall trends over time and to predict this year possible crop yield after concerning different factors.
- iv)We are also interested to examine different types of problems in Indian agricultural system and come up with ways to reduce these problems via policy measures or farmers outreach programs.

2 Introduction

2.1 Historical Background

India's agricultural heritage dates back to the Indus Valley Civilization, where sophisticated irrigation systems and granaries were already in use. Over centuries, agriculture in India evolved under various empires and rulers. During the colonial period, Indian agriculture suffered due to exploitative policies, leading

to famines and rural poverty. However, after independence in 1947, significant efforts were made to revive and modernize the sector.

The most transformative event in India's agricultural history was the Green Revolution in the 1960s and 70s by M. S. Swaminathan. With the introduction of high-yielding varieties (HYVs) of seeds, improved irrigation infrastructure, chemical fertilizers, and pesticides, food grain production, especially wheat and rice, saw a dramatic increase. It helped India achieve self-sufficiency in food production and reduced dependence on imports.

2.2 Present-Day Scenario

Today, agriculture in India is a vast and diverse sector. It includes the cultivation of crops, horticulture, dairy, fisheries, and forestry. The country is among the world's top producers of rice, wheat, sugarcane, cotton, fruits, and vegetables. According to recent statistics, agriculture and allied sectors contribute around 15-18 % to India's Gross Domestic Product (GDP) and employ nearly 50 % of the workforce.

India's agriculture is heavily dependent on monsoons, making it vulnerable to climate variability. Over 50 % of the net sown area is still rain-fed, despite an increase in irrigation coverage. In recent years, there has been a shift towards mechanization, organic farming, precision agriculture, and the use of agri-tech platforms to support farmers with weather forecasts, market prices, and modern techniques.

2.3 About major crops in India

Various food and non-food crops are grown in different parts of the country depending upon the variations in soil, climate, and cultivation practices. Major crops grown in India are rice, wheat, millet, pulses, tea, coffee, sugarcane, oil seeds, cotton, jute, etc. This can be classified into two classes:

- **Food crops**- Rice, Wheat, Millets, Maize and Pulses.
- **Cash crops**- Sugarcane, Oilseeds, Horticulture crops, Tea, Coffee, Rubber, Cotton and Jute.

Also based on the season we can classify into three classes :-

- **Kharif crops** [Sown: **June-July** Harvested: **September-October**]:- Rice (Aman), maize, jowar, bajra, tur, moong, urad, cotton, jute, groundnut, soybean etc.
- **Rabi crops** [Sown: **October-December** Harvested: **April-June**]- Rice (boro) , Wheat, barley, peas, gram, mustard etc.
- **Zaid crops** [Sown and harvested: **March-July** (between Rabi and Kharif)]- Rice(Aush) , Seasonal fruits, vegetables, fodder crops etc.

3 Analyzing the data

As said earlier , we are now going to identify those state which have highest production of different types of crops like paddy, wheat, jute etc. We have collected data about total production for each individual crop over the years for each state. Then we have sum the total production over the years for each crop. Then we make barplot to identify the highest production state. Also we plot the time series graph for each crop through all over the country to see the trend. It is to be noteworthy that the the years may vary for different crops due to the various availability of the data.

3.0.1 paddy

India is the world's second-largest producer of rice, and the largest exporter of rice in the world. We get a paddy production dataset of Indian states from 2013-14 to 2022-23. But the 2022-23 data is not complete. So we take care of the data from 2013-14 to 2021-22. We sum the total production over the years and then plot it:

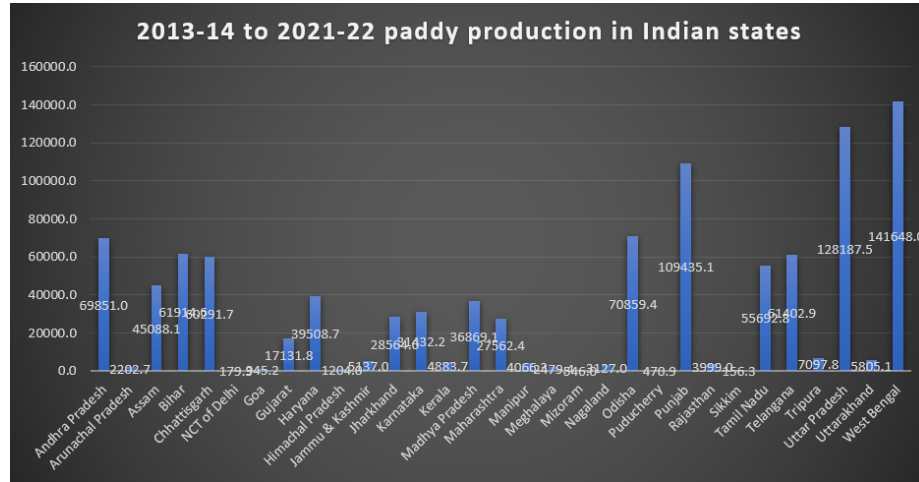


Figure 1: Paddy production of Indian states

so we get the top producers are:

First: West Bengal

Second: Uttar Pradesh

Third: Punjab

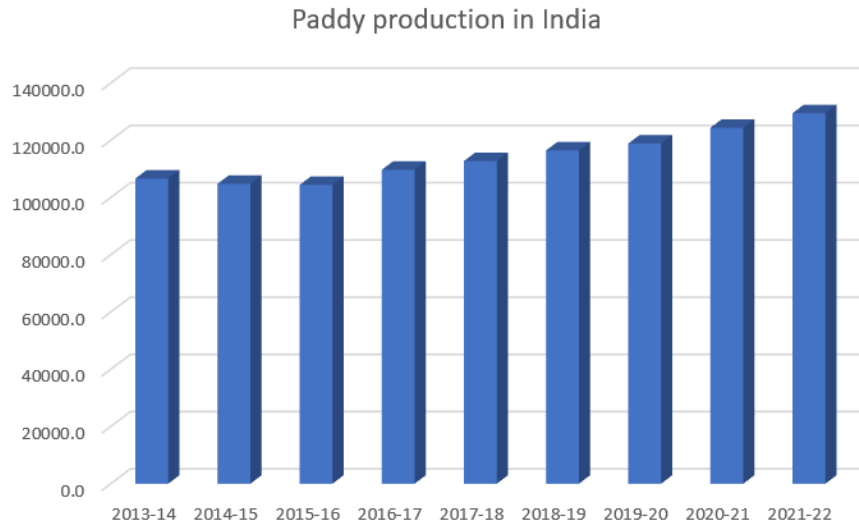


Figure 2: Paddy production of India

Now we are interested in the time series graph for all over India from 2013-14 to 2021-22.

So here we can see a good increasing trend over the years. That means the production for paddy is increasing day by day and it is also important for the increasing population.

3.0.2 wheat

Wheat is the second most important food crop. It is the main food crop in the north and north-western parts of the country. India is the second-largest producer of wheat. Here we get the state wise data over three years 2019-20 to 2022-23 and the all over India data from 2014-15 to 2024-25. The diagram is in the next page from which we get our three top producers states are:

First: Uttar Pradesh

Second: Madhya Pradesh

Third: Punjab

Note : This wheat data has some flaws. As we do not get any data regarding state wise production for the year 2014-15 to 2018-19 and after 2022-23 so we cannot do any further computation with this data. We just mention this data as wheat is the second important food crops of our country.

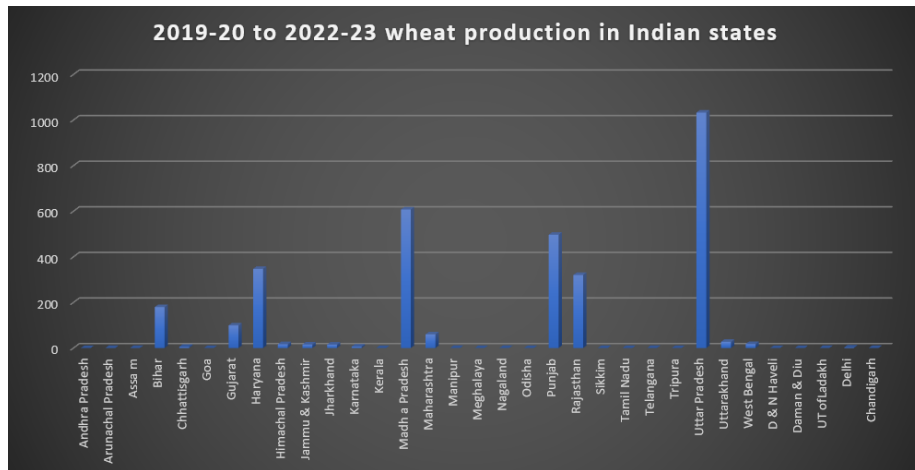


Figure 3: Wheat production of Indian states

Also have a look for all over the country.

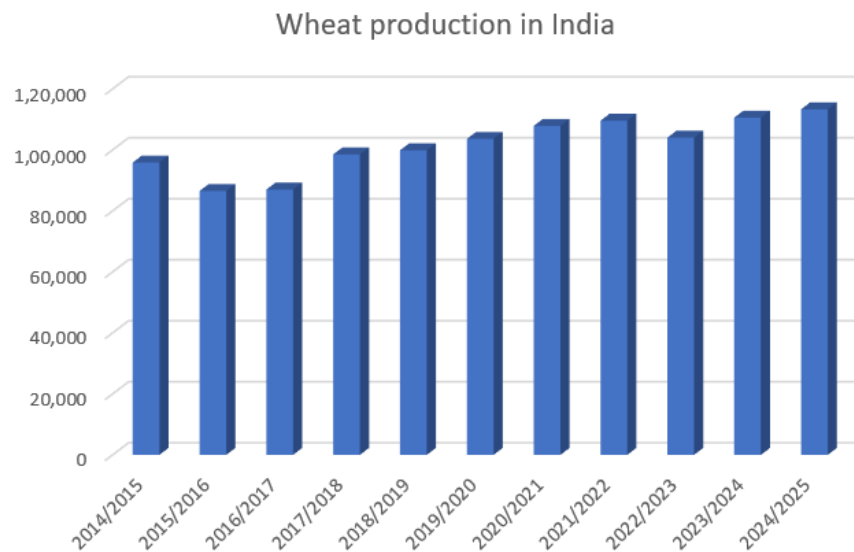
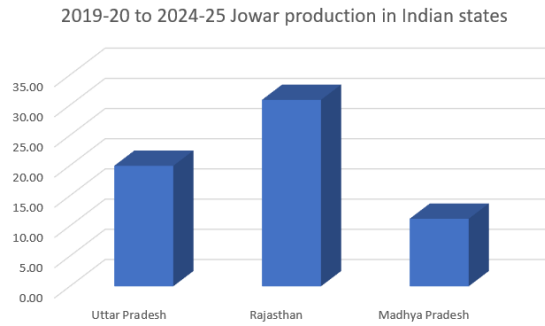


Figure 4: Wheat production of India

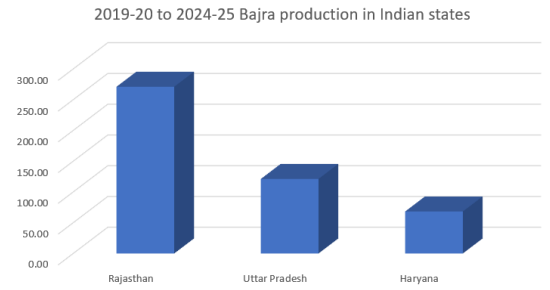
3.0.3 Other crops

As we have covered our main two food crops, we will give the production of top three states of India only, for the other crops. This is because the other crops are not growing all over the country and the data related to these production

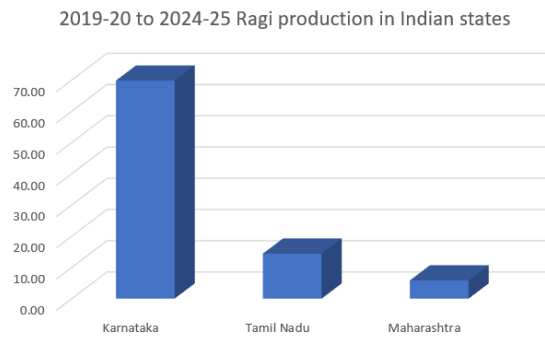
is not available as such. This all data are collected from *Unified Portal for Agricultural Statistics (UPAg)* . We have discussed some food crops (Jowar, Bajra, Ragi, Moong) and some cash crops(sugarcane,cotton,jute, mesta) one by one.



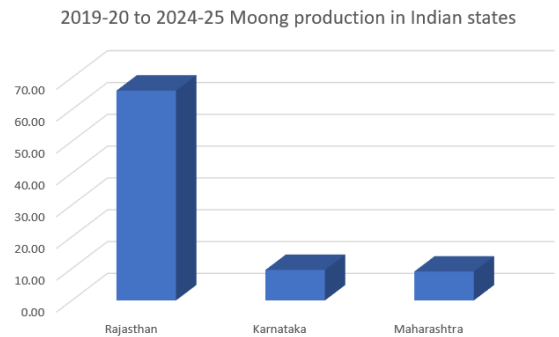
(a) Jowar production of Indian states



(b) Bajra production of Indian states



(a) Ragi production of Indian states



(b) Moong production of Indian states

So we get the top producers of these crops:

Jowar: Rajasthan

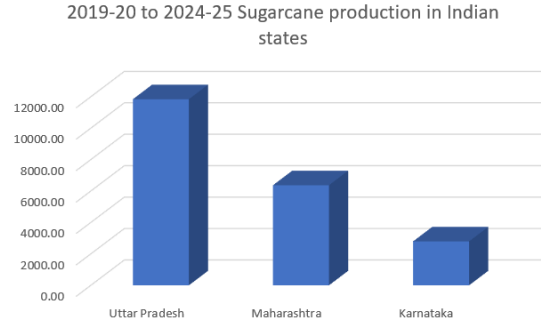
Bajra: Rajasthan

Ragi: Karnataka

Moong: Rajasthan

Now, let us discuss some cash crops:-

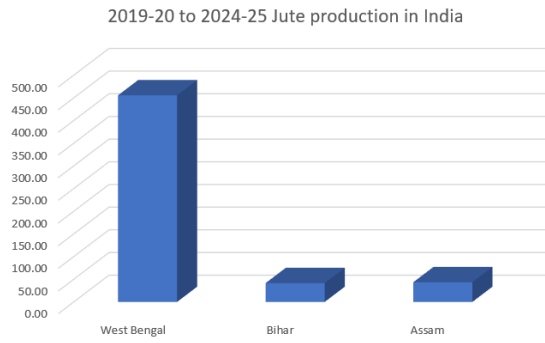
[note: - In the data set there are several other crops. But we cannot discuss all of them as they are not too much popular across the all over regions of India. Moreover this will just increase the length of this project. That's why we are only focusing on some of the crops and then want to see the production pattern and based on that want to predict the production value for future.]



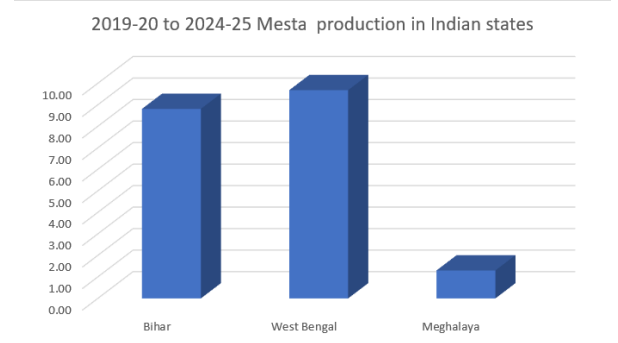
(a) Sugarcane production of Indian states



(b) Cotton production of Indian states



(a) Jute production of Indian states



(b) Mesta production of Indian states

So we get the top producers of these crops:

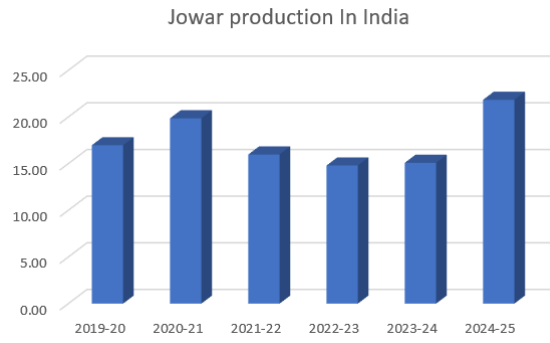
Sugarcane: Uttar Pradesh

Cotton: Maharashtra

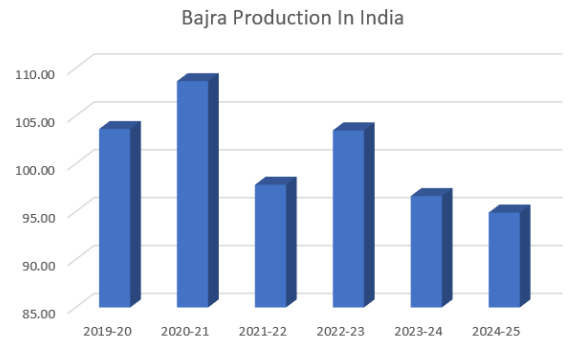
Jute: West Bengal

Mesta: West Bengal

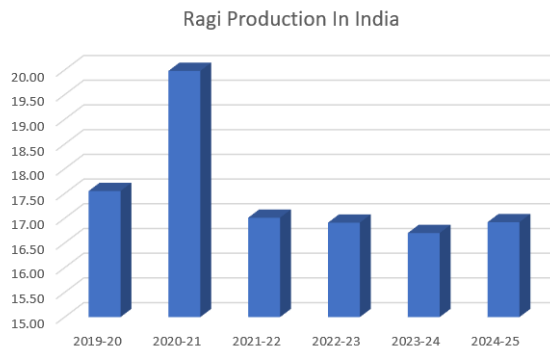
Now we discuss the overall India's trend for these crops over the years from 2019-20 to 2024-25. After seeing these trend we can see that the production for each year is not fixed obviously. We suspect that this variation is due to the climatic factor specially Rainfall and temperature. As these varies year to year it is our primary assumption that these factors have a crucial role in crop production. Hence, we want to find correlation among these variables. In the next section we have find partial correlation between crop production and temperature given rainfall and partial correlation between crop production and rainfall given temperature for the crops **paddy**, **Jowar** and **jute**. We omit the others as the other may show similar behavior and we do not want to extend this paper unnecessarily large.



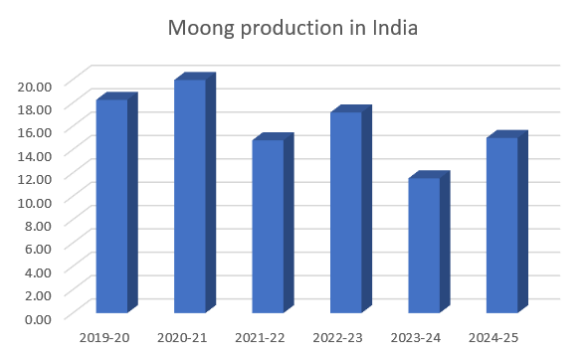
(a) Jowar production of India



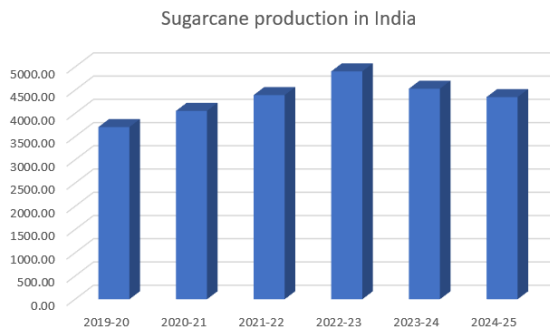
(b) Bajra production of India



(a) Ragi production of India



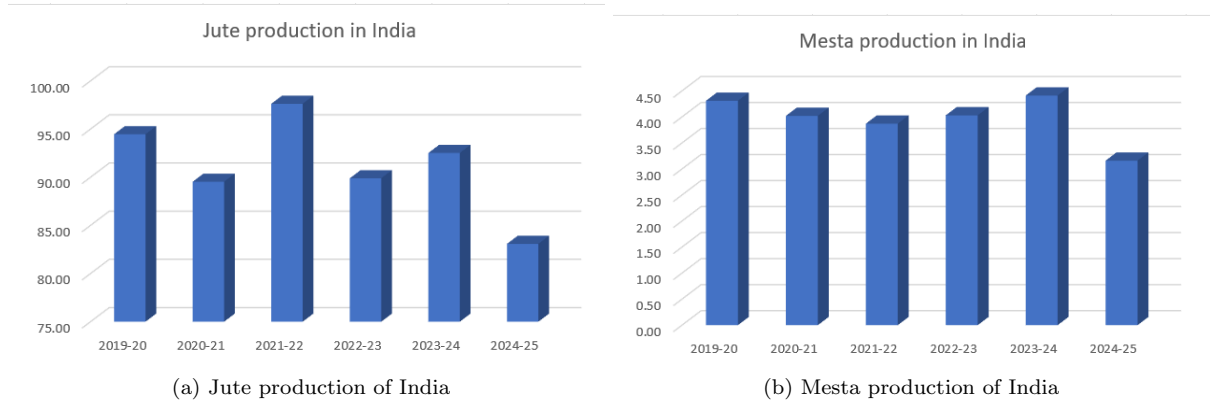
(b) Moong production of India



(a) Sugarcane production of India



(b) cotton production of India



3.1 Partial correlation

As said before we are now going to see the impact of Rainfall and Temperature on crop production. For this purpose we have computed partial correlation using R. The R codes are also given here:-

4 R code

Partial correlation between paddy production and temperature

Rahul Ganguly

2025-04-18

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 6 5

names(x)

## [1] "Rice" "Jowar" "Jute" "Rainfall" "Temp"

# Computing Partial correlation:-
f3 = lm(Rice-Rainfall, data = x) # Regress Rice on Rainfall
f4 = lm(Temp-Rainfall, data = x) # Regress Temp on Rainfall

f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f13,f14) # Partial correlation between Rice production and temp
r # Print required partial correlation

## [1] 0.163019
```

Figure 5: Partial correlation between Paddy production and temperature

Partial correlation between paddy production and rainfall

Rahul Ganguly

2025-04-18

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 6 5

names(x)

## [1] "Rice"    "Jowar"   "Jute"    "Rainfall" "Temp"

# Computing Partial correlation:-
f3 = lm(Rice-Temp, data = x) # Regress Rice on Temp
f4 = lm(Rainfall-Temp, data = x) # Regress Rainfall on Temp

f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f13,f14) # Partial correlation between Rice production and rainfall
r # Print required partial correlation

## [1] 0.1944514
```

Figure 6: Partial correlation between Paddy production and rainfall

Partial correlation between Jowar production and temperature

Rahul Ganguly

2025-04-18

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 6 5

names(x)

## [1] "Rice"    "Jowar"   "Jute"    "Rainfall" "Temp"

# Computing Partial correlation:-
f3 = lm(Jowar-Rainfall, data = x) # Regress Jowar on Rainfall
f4 = lm(Temp-Rainfall, data = x) # Regress Temp on Rainfall

f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f13,f14) # Partial correlation between Jowar production and temp
r # Print required partial correlation

## [1] -0.8394018
```

Figure 7: Partial correlation between Jowar production and temperature

Partial correlation between Jowar production and rainfall

Rahul Ganguly

2025-04-18

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 6 5

names(x)

## [1] "Rice"    "Jowar"   "Jute"    "Rainfall" "Temp"

# Computing Partial correlation:-
f3 = lm(Jowar-Temp, data = x) # Regress Jowar on Temp
f4 = lm(Rainfall-Temp, data = x) # Regress Rainfall on Temp

f3 = f3$residuals # Taking residuals(true value - estimated value) of f3
f4 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f3,f4) # Partial correlation between Jowar production and rainfall
r # Print required partial correlation

## [1] -0.8368093
```

Figure 8: Partial correlation between Jowar production and rainfall

Partial correlation between Jute production and temperature

Rahul Ganguly

2025-04-18

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 6 5

names(x)

## [1] "Rice"    "Jowar"   "Jute"    "Rainfall" "Temp"

# Computing Partial correlation:-
f3 = lm(Jute-Rainfall, data = x) # Regress Jute on Rainfall
f4 = lm(Temp-Rainfall, data = x) # Regress Temp on Rainfall

f3 = f3$residuals # Taking residuals(true value - estimated value) of f3
f4 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f3,f4) # Partial correlation between Jute production and temp
r # Print required partial correlation

## [1] 0.2810685
```

Figure 9: Partial correlation between Jute production and temperature

Partial correlation between Jute production and rainfall

Rahul Ganguly

2025-04-18

```
# Loading the data set:-
x = read.delim("C:/Users/rahul/OneDrive/Documents/proj4.txt")

# Knowing some primary things about the data set:-
dim(x)

## [1] 6 5

names(x)

## [1] "Rice" "Jowar" "Jute" "Rainfall" "Temp"

# Computing Partial correlation:-
f3 = lm(Jute-Temp, data = x) # Regress Jute on Temp
f4 = lm(Rainfall-Temp, data = x) # Regress Rainfall on Temp

f13 = f3$residuals # Taking residuals(true value - estimated value) of f3
f14 = f4$residuals # Taking residuals(true value - estimated value) of f4
r = cor(f13,f14) # Partial correlation between Jute production and rainfall
r # Print required partial correlation

## [1] 0.2613296
```

Figure 10: Partial correlation between Jute production and rainfall

So for each crop we can see some important observations:

Paddy: Partial correlation between paddy production and temperature given rainfall and Partial correlation between paddy production and rainfall given temperature are both positive. This imply that good amount of rainfall and temperature is perfect for paddy production. This seems to be true if we consider the scenario of West Bengal.

Jowar: Partial correlation between jowar production and temperature given rainfall and Partial correlation between jowar production and rainfall given temperature are both negative. This imply that dry weather is perfect for Jowar production. That's why the western part of India produce lot of millets than the other part of the country.

Jute: Partial correlation between jute production and temperature given rainfall and Partial correlation between jute production and rainfall given temperature are both positive. This again imply that good amount of rainfall and temperature is perfect for paddy production. Hence, West Bengal is in the top in jute production.

5 Future Prediction

Now we have arrived at the final part of this project. We want to find the future production of these crop using the previous data. We find the estimating

equation for the crops(paddy,jowar,jute) using time series analysis using excel. The scatterplot and corresponding fitted equations are shown below.

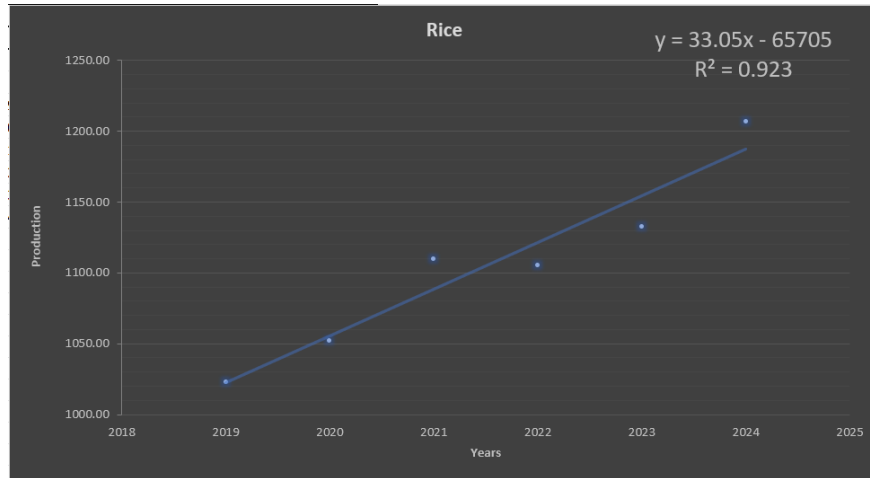


Figure 11: Rice production over the years (scatter plot and fitted curve)

So , we get the fitting equation is linear here. Using this $Y = 33.05x - 65705$ we find for 2025 the production will be = 1221.25 Lakh Tones.

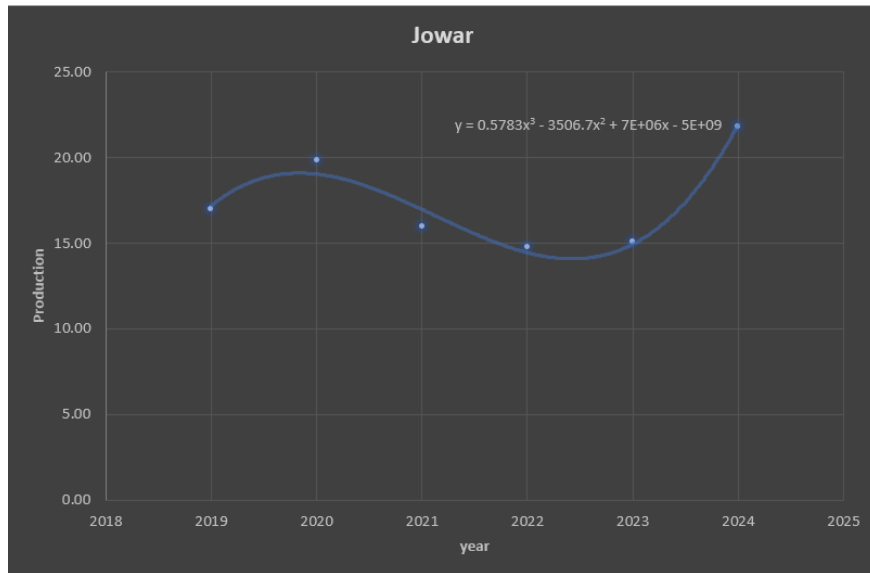


Figure 12: Jowar production over the years (scatter plot and fitted curve)

So , we get the fitting equation is cubic here. Using this $Y =$

$$0.5783x^3 - 3506.7x^2 + 7E + 06x - 5E + 09$$

we find for 2025 the production will be = 38.56 Lakh Tones.

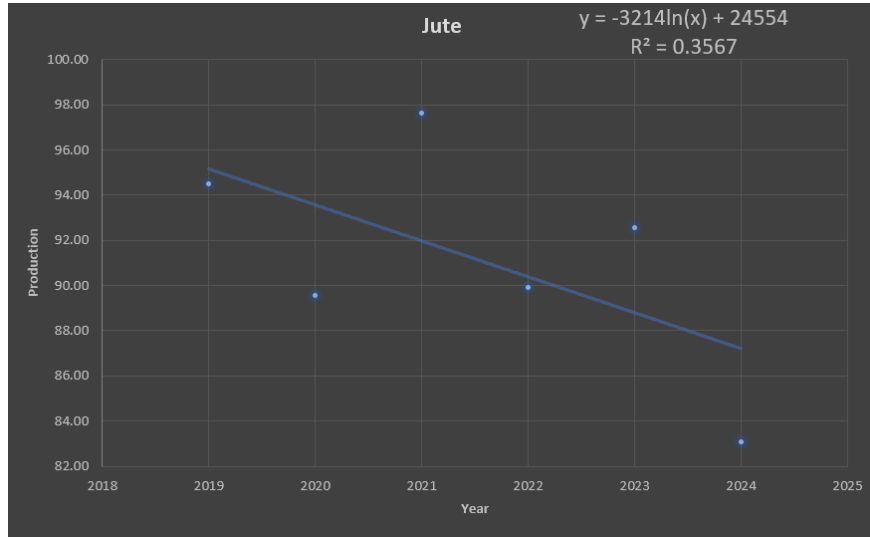


Figure 13: Jute production over the years (scatter plot and fitted curve

So , we get the fitting equation is logarithmic here. Using this $Y = -3214\ln(x) + 24554$ we find for 2025 the production will be = 84.77 Lakh Tones.

Proceeding in this way we can also analyze the other crops as well.

6 Problems in Agriculture in India and possible solutions(Government Roles):-

6.1 Problems

- i) The agricultural system in India is too much rainfall specifically monsoon dependent. The dry parts of India is not as good in agricultural activities.
- ii) The temperature also plays a crucial role in agriculture. The day by day temperature increasing i.e Global warming affect the agriculture.
- iii) In some of the crops we see that it's production is decreasing day by day. This means the farmers of that particular crop may loose interest for production due to low price of crop or high value of pesticides, fertilizers etc or in that region not such kind of high level technology are used.

6.2 Possible solution

Government have already taken several schemes like:-

1)**e-NAM (National Agriculture Market)**: For digital and unified market integration.

2)**Pradhan Mantri Fasal Bima Yojana (PMFBY)**: For crop insurance and risk mitigation.

3)**Soil Health Cards**: To promote judicious use of fertilizers.

4)**Agri-Infra Fund**: For post-harvest management and infrastructure.

5)**Farm Laws (now repealed)**: Though controversial, they aimed to liberalize agri-markets.

beside this government should also:

- i) inspire the farmers to harvest new crops.
- ii) Make institute to teach them about the use of new technologies.
- iii) Try to fix the market price for each crop so that they can get great energy and enthusiasm in their work.

7 Limitations of the project

i) The years we considered here are not same for every crop due to the unavailability of the data.

ii) In finding the correlation we have used the average temperature and rainfall data. But India is a large country, too much variation are seen between different regions of India. If in those cases we can use the local weather data that will give us more efficient result.

iii) During the calculation of future trend we are assuming that weather will be as before but with a large change in the amount of rainfall and temperature (though it has low chance of happening) it might affect the production.

8 Conclusion

Agriculture in India is more than just a means of livelihood—it is a way of life for millions and a cornerstone of the nation's socio-economic fabric. While the sector has shown resilience over the decades, it now faces the dual challenge of ensuring food security and improving farmers' income amid environmental, technological, and market transitions. The path forward lies in transforming agriculture from a survival activity to a sustainable and profitable enterprise through inclusive reforms, innovation, and investment. A future-ready Indian agriculture must be climate-smart, market-linked, and farmer-centric—only then can it truly become the engine of rural prosperity and national development.

9 Acknowledgment

I would want to convey my heartfelt gratitude to Prof. Ayanendranath Basu my mentor, for his invaluable advice and assistance in completing my project. He was there to assist me every step of the way, and his motivation is what enabled me to accomplish my task effectively. I would also like to thank him for assisting me by supplying the equipment that was essential and vital, without which I would not have been able to perform efficiently on this project.

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